

NEW RIVER MCAS, NC, USA

WMO: 723096

Lat: 34.708N

Lon: 77.440W

Elev: 26

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93727

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	26.5	5.8	7.4	31.1	10.3	9.3	33.7	24.6	46.2	21.9	46.6	5.9	10	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	92.7	77.8	90.6	77.3	88.7	76.5	80.4	88.0	79.3	87.0	78.3	85.4	8.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	148.8	84.1	77.2	141.8	82.8	76.2	136.9	82.0	43.9	88.5	42.7	86.9	41.7	85.8	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.4	16.4	16.6	97.0	3.8	2.1	13.9	98.5	11.6	99.7	9.5	100.9	6.8	102.4		
(5)			14.6	83.4	4.0	1.6	11.7	84.5	9.4	85.4	7.2	86.3	4.3	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	45.2	48.3	54.2	63.0	70.5	77.5	80.3	79.2	74.5	64.8	54.8	48.9		
	DBStd	14.47	10.57	9.56	9.02	7.68	6.24	4.62	3.38	3.51	4.90	7.62	8.47	9.65		
	HDD50	612	222	135	61	4	0	0	0	0	0	2	47	141		
	HDD65	2518	617	472	349	128	23	1	0	0	3	99	321	505		
	CDD50	5550	73	88	193	394	636	825	940	905	735	463	190	108		
	CDD65	1977	2	5	15	67	195	376	475	440	288	94	14	6		
	CDH74	16335	5	30	104	454	1418	3239	4523	3899	1995	589	63	16		
	CDH80	5504	0	4	12	84	393	1196	1764	1401	551	97	2	0		
	WSAvg	7.3	7.7	8.1	8.8	8.8	7.7	6.8	6.4	5.9	7.0	6.6	6.9	7.1		
	PrecAvg	53.9	4.1	4.1	4.0	3.1	4.2	5.2	7.4	6.3	5.4	2.9	3.0	3.6		
(15) Precipitation	PrecMax	70.1	8.3	10.7	8.0	8.9	11.5	11.8	14.0	12.8	22.4	8.9	7.1	6.6		
	PrecMin	38.2	1.4	1.3	0.7	0.5	0.6	2.2	3.5	0.6	0.9	0.6	0.6	0.4		
	PrecStd	7.2	2.0	2.3	1.8	1.9	2.5	2.2	2.6	3.3	4.3	1.9	1.7	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.3	78.4	81.7	86.3	91.5	94.8	95.4	94.9	90.3	86.2	79.4	76.3		
		MCWB	64.6	65.8	65.0	67.6	74.3	77.3	79.0	78.9	75.6	73.2	68.3	66.9		
	2%	DB	70.2	72.4	76.8	82.3	87.7	91.7	92.8	91.3	87.9	82.4	75.5	72.4		
		MCWB	63.0	62.0	63.2	67.0	73.2	76.8	78.4	78.2	74.9	71.7	65.8	65.5		
	5%	DB	66.3	68.2	73.0	79.4	84.1	89.3	90.7	89.3	85.6	80.0	72.6	68.9		
		MCWB	60.8	60.4	61.1	65.9	71.8	76.2	78.1	77.7	74.1	70.0	64.9	64.0		
	10%	DB	62.9	64.3	69.7	75.9	81.5	86.7	88.4	87.4	83.3	77.3	69.7	65.6		
		MCWB	57.7	57.6	60.0	64.3	70.1	75.1	77.2	76.6	73.4	69.2	63.6	61.5		
	0.4%	WB	67.5	68.0	68.6	73.6	78.2	80.6	82.1	82.3	79.0	77.3	72.2	70.2		
		MCDB	70.6	74.2	75.3	80.0	85.9	89.2	90.8	89.3	85.0	80.7	76.3	73.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	64.9	65.6	66.5	70.6	75.5	79.0	80.4	80.4	77.7	75.2	69.5	67.2		
		MCDB	68.6	70.4	72.5	77.2	83.1	87.9	88.6	87.2	83.5	79.9	72.7	70.4		
	5%	WB	62.0	62.5	64.4	68.6	73.9	77.7	79.4	79.1	76.6	73.2	67.0	64.7		
		MCDB	65.7	66.3	69.9	75.4	80.9	86.0	87.4	86.0	82.1	77.7	70.7	68.0		
	10%	WB	58.5	58.7	62.2	66.8	72.4	76.5	78.3	78.0	75.5	70.9	64.3	61.7		
		MCDB	62.2	63.9	67.9	73.4	78.9	83.8	85.7	84.6	80.7	75.3	68.3	65.5		
	5% DB	MDBR	19.9	20.9	21.6	21.0	19.0	17.1	16.1	15.8	16.0	19.6	21.5	19.8		
		MCDBR	23.5	24.6	24.6	23.1	20.6	19.6	18.2	17.8	18.0	20.2	23.1	21.6		
		MCWBR	17.3	15.9	13.6	11.6	9.3	8.2	7.5	7.7	8.0	10.3	14.8	15.5		
		MCWBR	20.8	22.3	20.2	18.8	17.7	17.7	16.5	15.8	14.6	15.4	19.2	19.9		
(40) Clear-Sky Solar Irradiance	taub		0.322	0.332	0.362	0.399	0.451	0.493	0.525	0.505	0.423	0.379	0.344	0.328		
		taud	2.524	2.491	2.429	2.318	2.218	2.148	2.073	2.124	2.354	2.452	2.504	2.539		
	Ebn at Noon		281	290	289	281	267	255	246	248	266	270	270	272		
		Edh at Noon	26	30	35	41	45	49	52	49	37	31	26	24		
(44) All-Sky Solar Radiation	RadAvg		846	1075	1417	1779	1951	2023	1915	1738	1446	1228	936	749		
	RadStd		53	101	108	112	134	110	110	103	99	135	68	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+281	+166
(47) Regional (40 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	+286	+207

Nomenclature: See separate page